

PANASHE S. BOBOJANI

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING GRADUATE | SOFTWARE ENGINEER

Harare, Zimbabwe · +263 78 069 6934 · panasheshaunbobojani@gmail.com

PROFESSIONAL SUMMARY

AI and Machine Learning graduate (BSc Honours, University of Zimbabwe) with practical software engineering experience spanning machine learning, full-stack development, and public-sector digital transformation. Built and shipped systems ranging from an AI-powered document intelligence platform (OCR, NLP, predictive classification) to full-stack web applications using Python, React, and Firebase, including automation-focused ICT work during a nine-month placement at Zimbabwe's Ministry of Information Communication Technology, Postal and Courier Services. Combines a strong academic foundation in machine learning, deep learning, natural language processing, data mining, and database systems with hands-on delivery experience and the ability to translate technical work for non-technical stakeholders. Seeking an AI Engineer, Machine Learning Engineer, Software Engineer, Data Analyst, ICT Officer, or Graduate Trainee role.

CORE COMPETENCIES

End-to-End Software & ML Solution Delivery • Applied Machine Learning, Deep Learning & NLP • Full-Stack Development (Python, React, Firebase) • Data Analysis, Mining & Insight Generation • Digital Transformation & Process Automation • Technical Communication & Stakeholder Training • Cross-Functional Collaboration • Rapid Technical Learning & Adaptability

TECHNICAL SKILLS

Programming Languages Python · JavaScript/TypeScript · SQL · C#

AI & Machine Learning TensorFlow · PyTorch · scikit-learn · XGBoost · Deep Learning (CNN, LSTM, MLP) · NLP (NER) · Computer Vision · Predictive Modelling

Data & Analytics Pandas · Excel · Data Mining · Recommender Systems · Data Cleaning & Analysis · OCR

Software Development React · FastAPI · HTML/CSS · Unity · Jinja2

Cloud & Databases Firebase · Firestore · SQL Databases

Tools & Platforms Git · Google Colab · VS Code

PROFESSIONAL EXPERIENCE

Industrial Attachment — Applications Development, AI Team & Digital Skills Training

Ministry of Information Communication Technology, Postal and Courier Services, Harare — Nov 2024 – Aug 2025

- Developed and maintained internal ministry applications using Python, C#, Java and HTML, with Firebase-backed data handling, supporting digital operations within the Applications Development and Management department.
- Engineered automation-focused AI initiatives, applying machine learning techniques to operational data as part of the ministry's data-driven decision-support efforts.
- Delivered digital skills training to stakeholders, translating technical concepts into accessible material and strengthening cross-audience technical communication.
- Worked across three functional teams — Applications Development, AI Initiatives, and Digital Skills Training — gaining direct exposure to software delivery, applied AI, and digital transformation within a government ICT environment.
- Operated at the intersection of ICT policy, systems delivery, and institutional stakeholder needs, deepening understanding of technology adoption at scale.

SELECTED PROJECTS

InsureIntel Zimbabwe — AI-Powered Insurance Document Intelligence Platform (Capstone)

Problem: Insurance regulatory compliance in Zimbabwe requires manual review of unstructured documents against IPEC FSR-1 reporting requirements — slow and inconsistent at scale.

Technologies: OCR, Named Entity Recognition (NLP), XGBoost classifier, rules-based compliance engine, Jinja2 admin dashboard.

Technical Complexity: Combined OCR and NLP for unstructured information extraction with a machine learning classifier for document categorisation, then layered a compliance engine on top to validate output against IPEC FSR-1 requirements.

Business Value: Gives compliance teams an automated first pass on regulatory document review, cutting reliance on manual reading of unstructured paperwork.

AutoDirect — Freight Logistics Coordination Platform

Problem: Freight operators needed one system to manage the full lifecycle of logistics jobs instead of fragmented manual coordination.

Technologies: Python, React, Firebase, Cloud Functions, Firestore.

Technical Complexity: Engineered a Cloud Functions orchestration layer for job lifecycle logic, then re-architected it around client-side Firestore transactions to operate within Firebase's free-tier constraints — a deliberate cost/architecture trade-off.

Business Value: Gives freight operators one system to coordinate jobs instead of relying on scattered manual processes.

3D Smart Document Directory (In Development)

Problem: Flat, folder-based directories scale poorly for visually exploring large, classified document sets.

Technologies: Python, document classification and clustering techniques, 3D interface design.

Technical Complexity: Extending prior clustering and information-retrieval work into an interactive, spatially navigable interface for document classification and retrieval.

Business Value: Aims to make large document collections easier to explore visually rather than through flat lists — currently in active development.

Car Service Company Web App

Problem: A vehicle servicing business needed to manage bookings and service tracking online instead of manually.

Technologies: Python, React, Firebase.

Technical Complexity: Built a full-stack booking and service-management application covering customer bookings, service tracking, and admin oversight.

Business Value: Streamlines scheduling and service tracking for the business and its customers.

EDUCATION

BSc Honours, Artificial Intelligence & Machine Learning

University of Zimbabwe — 2022–2026

Degree Classification: 2.1 (Upper Second Class Honours) — Capstone: InsureIntel Zimbabwe (AI-powered insurance document intelligence platform)

Relevant Coursework: Machine Learning & Neural Networks, Deep Learning (CNN, LSTM, MLP), Natural Language Processing, Artificial Intelligence & Expert Systems, Data Mining & Recommender Systems, Software Engineering, Database Design & Development, Big Data Technologies.

Academic Projects: Built a CNN-based chest X-ray pneumonia classifier, an LSTM stock-price prediction model, and multiple recommender-system labs (cold-start design, SVD diagnostics, feedback-loop detection) as coursework deliverables.

Advanced Level — St Francis of Assisi, Chivhu (2020–2021): Mathematics, Computer Science, Physics

Ordinary Level — St Francis of Assisi, Chivhu (2016–2019): 7 A's, 3 B's, 2 C's, 1 Distinction

CERTIFICATIONS

Scrum Fundamentals Certified

Six Sigma Yellow Belt

ACHIEVEMENTS

- Completed 4 Alison Secure Software courses, strengthening secure development practice.
- Progressed to mock-exam level in Huawei's HCIA IoT & Physical Computing and HCIA Data Communications & Networking programmes.
- Entrusted with digital skills training delivery to stakeholders during a government ICT attachment — a responsibility beyond standard intern duties, reflecting communication ability and institutional trust.

ADDITIONAL INFORMATION

Languages: English (Fluent), French (Intermediate)

Interests: Chess, Sports Analytics, Music

References available on request.